

Homework 9

Due April 12th

Problems:

1. A non-inverting amplifier (R_1 is input resistor, R_2 is feedback resistor) is built with a gain bandwidth product of 10 MHz and a non-zero output resistance $r_o = 75 \Omega$ (assume $r_d = \infty$). What is the maximum purely capacitive load, C_L , that could be added to the output such that $\phi_M \geq 45^\circ$ for:
 - (a) $R_1 = R_2 = 20 \text{ k}\Omega$,
 - (b) $R_1 = R_2 = 200 \text{ k}\Omega$,
 - (c) $R_1 = 2 \text{ k}\Omega$, $R_2 = 18 \text{ k}\Omega$, and
 - (d) $R_1 = \infty$, $R_2 = 0 \Omega$ (voltage follower)
2. Verify your answers to question 1 using LTspice: simulate and measure $T(jf)$ for each part. It is suggested to use UniversalOpAmp1 and set the GBP accordingly. Note that this model does not include r_o ; therefore, an external resistor must be added to model the output resistance.
3. In LTspice: using the same circuit in question 1 and $R_1 = R_2 = 20 \text{ k}\Omega$, let $C_L = 1 \mu\text{F}$, resulting in a very small ϕ_M (note, since there are only two poles in this circuit, it will technically never go unstable). Modify the circuit such that $\phi_M \approx 45^\circ$ by:
 - (a) Implementing a snubber network (resistor+capacitor in parallel with C_L) and tuning as described the lecture 17 notes, and
 - (b) Adding a feedback capacitor, C_f , and output series resistor, R_s , (also shown in lecture 17 notes). Assume that $R_s = 1 \Omega$.

Comment on the expected low-frequency gain error and gain peaking and/or oscillatory behavior of each system. Note that this can be done by simply interpreting the simulated $T(jf)$ but could also be simulated via a transient analysis with step input (akin to default design project input).
4. -6050 Only: Book Problem 8.50 (a+b only): This is more a qualitative problem than a quantitative one. Assume a generic op amp (such as the 741 amps: GBP = 1 MHz, $f_b = 5 \text{ Hz}$, and $a_o = 200 \text{ V/mV}$). You may also assume either:
 - that the crossover frequency, f_x , occurs at (1) a higher frequency than any/all poles and zeros within the feedback network and (2) $f_x > f_b$;
 - OR, use component values:
 - Fig. 3.11: $R_1 = R_2 = 10 \text{ k}\Omega$, $C_1 = 1 \mu\text{F}$, $C_2 = 10 \text{ nF}$.
 - Fig. 3.32: $R_1 = 1 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$, $R_3 = 100 \text{ k}\Omega$, $C_1 = 100 \text{ nF}$, $C_2 = 10 \text{ nF}$
5. Book Problem 8.55.